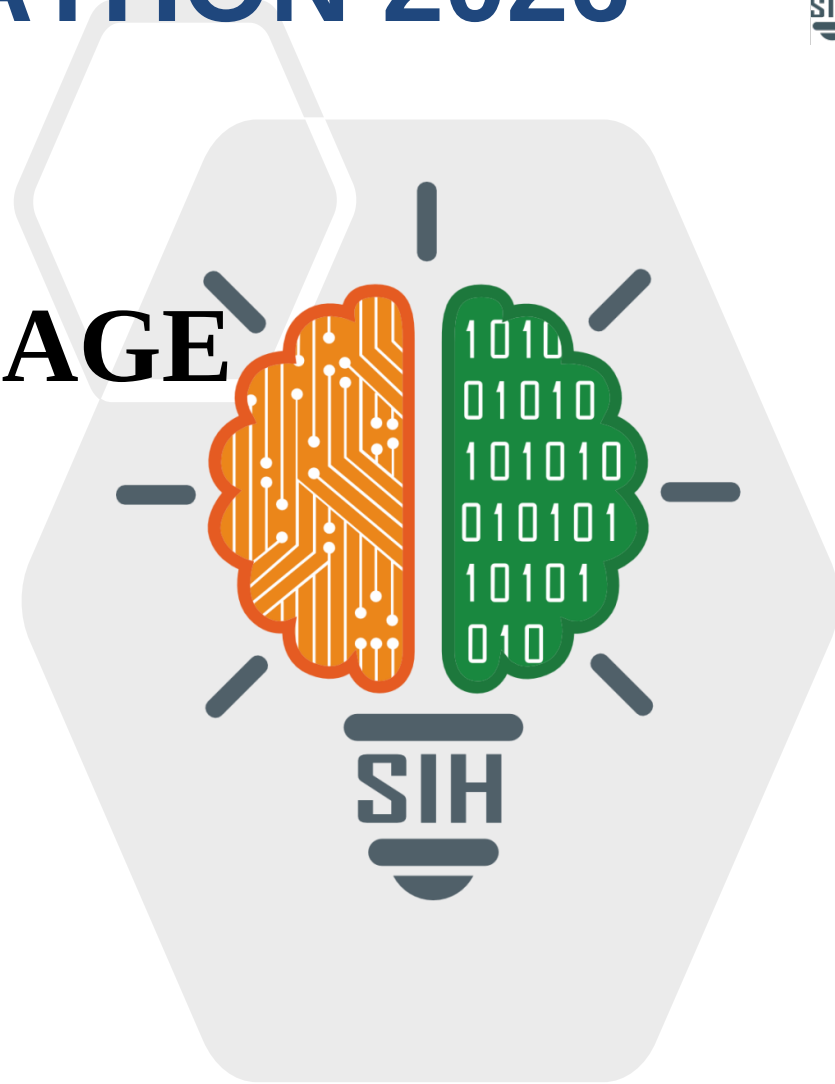


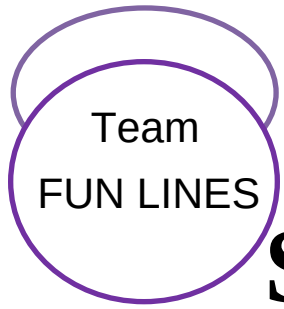
SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID – SIH26085
- Problem Statement Title – Smart River Water Level and Quality Surveillance
- Theme – AI / IoT / Environment
- PS Category – Software
- Team ID –
- Team Name – FUN LINES

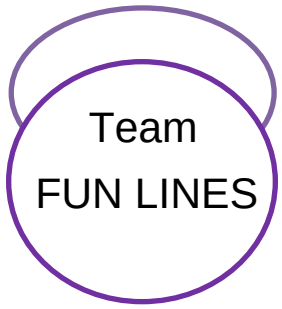




SMART RIVER SURVEILLANCE

Proposed Solution (Describe your Idea/Solution/Prototype)

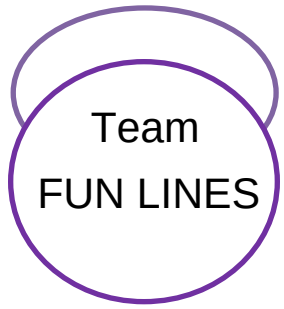
- **Detailed Idea:** Live GIS web portal syncing 40+ river stations instantly via CPCB API.
- **Problem Addressed:** Replaces 3-day lab delays with sub-second river health monitoring and early flood alerts.
- **Innovation:** Serverless edge proxy, offline-resilient caching, and automated 1-click A4 PDF inspection reports.



T TECHNICAL APPROACH



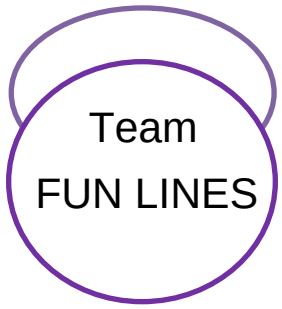
- **Technologies:** HTML5, Leaflet GIS map, Chart.js, Node.js Proxy on Vercel Edge, Express API.
- **Methodology:** Ingestion (Telemetry) → Normalization (CORS/Proxy) → WQI Computation Engine → GIS Sync → Automated PDF Reporting.



FEA FEASIBILITY AND VIABILITY



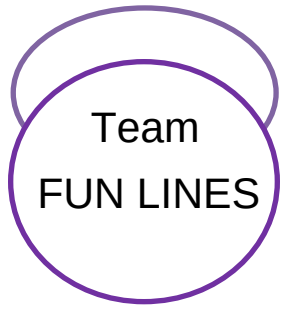
- **Analysis:** Highly feasible; 100% working prototype tracking Indian rivers with zero monthly server costs.
- **Challenges:** Govt server CORS blocking, telemetry API network timeouts, and raw data outlier spikes.
- **Strategies:** Edge proxy middleware, multi-tier resilient fallback caching, and mathematical bounds validation.



I IMPACT AND BENEFITS



- **Target Audience:** CPCB/SPCB pollution officers, disaster flood management units, and communities.
- **Social Benefit:** Prevents water-borne disease outbreaks before toxic water enters municipal pipelines.
- **Economic & Environmental Benefit:** Saves crores on manual testing; protects aquatic biodiversity.



RESE RESEARCH AND REFERENCES



- **CPCB RTWQMS Telemetry:** rtwqmsdb1.cpcb.gov.in
- **CPCB Live Webcam Network:** rtwqmsdb1.cpcb.gov.in/images/stations/
- **Methodology:** Brown et al. 'Weighted Arithmetic WQI Method'
- **Live Prototype:** smart-river-surveillance-project.vercel.app